

■ 2.7 Advanced Topic: The Naming of Things

■ 2.7.1 Aliases

It is conventional in *Mathematica* to name objects as explicitly as possible, typically using full English words. This means that names in *Mathematica* can sometimes get quite long.

If you use a particular long name very often, you may want to define a shorter “alias” for it. Especially in writing programs, however, the extra clarity you get by using an explicit, long, name much more than offsets any reduction in time or space that you might get from using a shorter alias.

<code>input ::= translation</code>	specify a translation for an input symbol name
<code>Alias["input"]</code>	show any translation that has been defined for <i>input</i>
<code>Alias[]</code>	show all translations that have been defined
<code>"input" ::= .</code>	remove a translation for <i>input</i>
<code>UnAlias[]</code>	remove all translations

Functions for defining aliases.

This defines `int` to be an alias for `Integrate`.

```
In[1]:= int ::= Integrate
Out[1]= Integrate
```

You can now type `int` instead of `Integrate`. What you gain in typing speed by doing this is usually not worth what you lose in clarity.

```
In[2]:= int[Sin[x]^3, x]
Out[2]= 
$$\frac{-3 \cos[x]}{4} + \frac{\cos[3 x]}{12}$$

```

Setting up an alias with `int::=Integrate` is in some respects like setting up a transformation rule with `int:=Integrate`. There is an important difference, however. An alias set up with `int::=Integrate` is used *wherever* `int` appears, whether it is evaluated or not. A transformation rule set up with `int:=Integrate` is only used, however, when `int` is *evaluated*.

This gives the translation for `int` that you have defined.

```
In[3]:= Alias["int"]
Out[3]= Integrate
```

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This removes the translation for `int`.

```
In[4]:= "int" :=.  
Out[4]= int
```

This sets up `aa` as an alias for `Aardvark`.

```
In[5]:= aa := Aardvark  
Out[5]= Aardvark
```

Whenever you type `aa`, it is immediately replaced by `Aardvark`. You can thus use `aa = value` to define a value for `Aardvark`.

```
In[6]:= aa = 6  
Out[6]= 6
```